



Descriptive Statistics

Reading a Distribution: Centre, Spread, Symmetry, Tails

Mean · Median · Standard Deviation · Skewness · Kurtosis

A working session for analysts who need to turn a column of numbers into a **defensible, actionable statement**. By the end, you will know not just how to compute each statistic — but which one is honest for the data in front of you.

Why This Session Exists

Every dataset you will ever open is a column of numbers you cannot read one row at a time. A 5,000-row order table is unreadable. Five numbers describing it are not. **Descriptive statistics are the compression layer between raw data and a decision.**

The Real Risk

The risk is not that you compute them wrong. The risk is that you **compute the wrong one** — and it still looks perfectly reasonable on the surface.

What This Session Covers

Half of this session is formulas. The other half is knowing which formula is **honest for the data in front of you** — and which one quietly lies.

A Simple Test

Ask yourself: what does "average salary" mean on a job portal? If you and a colleague disagree, that disagreement is the entire point of this session.

The Four Questions a Distribution Answers

Any single numeric column can be interrogated on exactly **four dimensions**. Answer all four and you have described the data. Answer only the first and you have described almost nothing — yet that is what most analysts stop at.

Dimension	What It Asks	Statistics
Centre	Where do values sit?	Mean, Median, Mode
Spread	How far apart are they?	Range, IQR, Variance, SD
Symmetry	Is one side longer than the other?	Skewness
Tails	How extreme are the extremes?	Kurtosis

📌 These four questions form the backbone of every analysis in this session. We will return to them repeatedly using the same dataset.

The Running Case: Meridian Retail

We will use **one dataset for every calculation** in this session, so the numbers stay comparable and the impact of each statistic is visible across the full arc of the analysis.

The Data

Order values from a single store day (₹ thousands), **n = 9**:

12, 14, 15, 15, 18, 20, 22, 25, 99

What's in It

Eight ordinary retail orders and **one bulk corporate order**. Realistic, messy, and deliberately structured to break every naive statistic.

Why These Numbers

Small enough to compute by hand.
Structured enough to expose the limits of mean, SD, and z-scores. Every formula in this deck will be re-derived on this one row.

Keep these nine numbers in mind throughout. Whenever a statistic shifts dramatically by removing the value of 99, that shift is itself the lesson.

Population vs. Sample

The formulas change depending on what you are holding. This distinction is not academic — it determines which version of every formula is correct for your situation.

Population

Every unit that exists in the group you care about. Example: all 40,000 Meridian orders in FY25.

Described by **parameters**, written with Greek letters: μ (**mean**), σ (**SD**).

Sample

The subset you actually measured. Example: the 9 orders from one store day.

Described by **statistics**, written with Latin letters: $\bar{x}$ (**mean**), s (**SD**).

You almost always hold a sample and want to say something about the population. **That gap is why one formula divides by n and another by $n - 1$.**

Data Types Decide Which Statistic Is Legal

Not every statistic can be computed on every column — even when the software allows it. Pandas will happily average a column of pin codes. The output is a number. It is meaningless. The rule is simple: **if the difference between two values has no unit, the mean has no interpretation.**

Data Type	Example	Mean	Median	Mode	SD
Nominal	Payment method, city	No	No	Yes	No
Ordinal	Satisfaction 1–5	Debatable	Yes	Yes	Debatable
Interval / Ratio	Order value, age	Yes	Yes	Yes	Yes

Always audit the column type before choosing a statistic. The data type sets the ceiling on what is analytically legal.

Notation Reference

Setting this down once so every formula later reads cleanly. These symbols appear in every statistics textbook, every Python output, and every academic paper you will read as an analyst.

1

Observations

x_i — the i -th individual observation in your dataset. So x_1 is the first, x_5 is the fifth, and so on.

2

Count & Summation

n — number of observations in the sample. Σ — sum over all observations from $i=1$ to n .

3

Centre Symbols

$\bar{x}$ — sample mean. μ — population mean. Both describe the centre; one is an estimate, one is the truth.

4

Spread Symbols

s — sample SD. σ — population SD. s^2 and σ^2 are the corresponding variances.

PART 1 — THE MEAN

Centre by Balance Point



The Mean: Definition

The arithmetic mean is the **total distributed equally across all units**. It answers the question: if every observation were identical, what would each one be?

The Formula

$$\bar{x} = (\sum x_i) / n$$

Sum all values, divide by the count.
Deceptively simple.

Its Greatest Strength

The mean **uses every single data point**. It is the only measure of centre with a clean algebraic relationship with the total: mean $\times$ n = sum. This makes it essential for any calculation that needs the total to be recoverable.

Its Greatest Weakness

Because it uses every data point, **every data point pulls on it** — including those that are far from the rest. A single extreme value can move the mean substantially, making it a poor descriptor of a typical observation.

Half the punchlines in this session depend on remembering that the mean uses every data point. Keep that property in mind as we proceed.

Mean — Worked Calculation

Using Meridian Retail order values (₹ thousands): **12, 14, 15, 15, 18, 20, 22, 25, 99**

With All 9 Orders

$$12 + 14 + 15 + 15 + 18 + 20 + 22 + 25 + 99 = \mathbf{240}$$

$$\bar{x} = \mathbf{240 / 9 = 26.67}$$

Notice that 26.67 is higher than 7 of the 9 actual order values. The mean is already not representative of most transactions.

Without the Bulk Order

$$12 + 14 + 15 + 15 + 18 + 20 + 22 + 25 = \mathbf{141}$$

$$\bar{x} = \mathbf{141 / 8 = 17.63}$$

One observation out of nine moved the reported average by **51%**. That single corporate order — which is a real, valid transaction — nearly doubles what the "average order" appears to be.

 A 51% shift from one observation is not a curiosity. It is a warning that the mean alone cannot describe this data honestly.

Two Properties of the Mean That Matter

These two algebraic properties are not trivia. They generate almost everything else in this session — variance, standard deviation, and the reason the median exists as an alternative.

Property 1: Deviations Sum to Zero

$\sum (x_i - \bar{x}) = 0$, always, for every dataset without exception.

This is why we cannot use raw deviations to measure spread — they cancel each other out perfectly. Positive deviations and negative deviations are exactly balanced. This forces us to square them when building a measure of spread, which is the origin of variance and standard deviation.

Property 2: The Mean Is Not Resistant

Every observation pulls on the mean **in proportion to its distance** from the centre. A value far from the rest pulls hard. A value close to the centre barely moves it.

This is the algebraic price of using every data point. The mean is efficient with clean data and badly behaved with dirty data. There is no version that does both.

The Weighted Mean

Used whenever observations do not carry equal importance. In business, this is far more common than the simple average.

Formula: $\bar{x}_w = \frac{\sum(w_i \cdot x_i)}{\sum w_i}$

Product Line	Revenue (₹ Cr)	Gross Margin	Weight
Electronics	120	8%	30%
Apparel	40	32%	10%
Grocery	240	4%	60%

Simple Average of Margins

$(8 + 32 + 4) / 3 = 14.7\%$

Each product line gets equal weight. But Electronics and Apparel together are less than half the revenue of Grocery alone.

Revenue-Weighted Margin

$(120 \times 8 + 40 \times 32 + 240 \times 4) / 400 = 8.0\%$

The unweighted number **overstates profitability by nearly 2×**. Portfolio reporting, index construction, and blended-rate calculations are all weighted means. Use them.

Where the Mean Breaks

The mean fails when the distribution has a long tail — which describes **most commercial data**. In all of these domains, "average" is a number that describes almost nobody in the actual dataset.



Income & Salary

A small number of very high earners pull the mean far above what most employees actually earn. Median salary is the honest number.



Insurance Claims

Most claims are small, routine, and affordable. A handful of catastrophic claims hold the financial risk. The mean claim blends them unhelpfully.



Customer Lifetime Value

A thin band of high-value customers generates a disproportionate share of revenue. Averaging them in inflates what a "typical" customer looks like.



API & Page Load Latency

Most requests are fast. A few are catastrophically slow. The mean latency tells you nothing about the user experience of the slowest requests.

The Trimmed Mean

A controlled compromise: keep the structure of the mean, but discard the most extreme observations before computing it.

01

Sort the Data

Arrange all observations in ascending order.

02

Remove a Fixed Percentage from Each End

Trim equally from the bottom and the top. A 10% trim removes the bottom 10% and the top 10% of observations.

03

Average the Remainder

Compute the arithmetic mean on what remains.

Meridian Example

Trim one observation from each tail: **14, 15, 15, 18, 20, 22, 25**

Trimmed mean = $129 / 7 = \mathbf{18.43}$ vs untrimmed **26.67**

Used In Practice For

- Olympic and sports scoring (drop the highest and lowest judge)
- CPI (Consumer Price Index) construction by statistical agencies
- Sell-side consensus earnings estimates
- Any benchmark where a single submission should not move the aggregate

Trade-off: You are discarding real observations. Always declare the trim percentage when reporting a trimmed mean.

When to Report the Mean

The mean is a powerful statistic — but only in the right conditions. Use this checklist before leading with it.

✓ Report It When

- The distribution is roughly symmetric
- You need the total to be recoverable ($\text{mean} \times n = \text{sum}$)
- You will feed the number into further algebra — variance, regression, and ANOVA all build on the mean
- Every observation genuinely deserves equal weight

✗ Avoid It When

- The tail is long and the audience will read the number as "typical"
- The data is bounded at zero with no upper bound (most commercial quantities)
- You are reporting to a public or non-technical audience who will interpret the number intuitively
- The outliers are real, valid, and numerous

📌 When in doubt: report both mean and median. Let the gap between them tell the story.

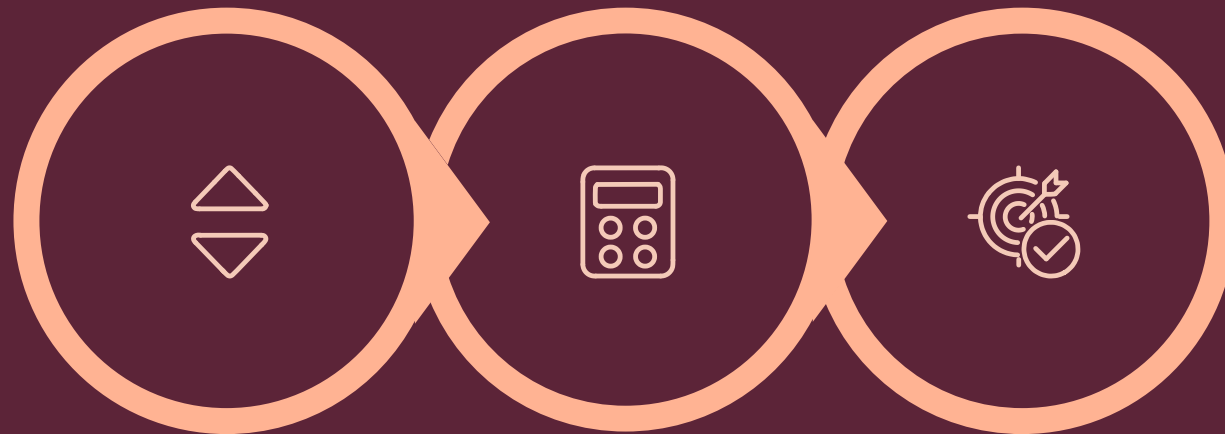
PART 2 — THE MEDIAN

Centre by Position



The Median: Definition

The median is the value at the **50th percentile** — half the observations fall below it, half above. Unlike the mean, it ignores the magnitude of the values above and below it. It only counts **positions**.



Sort Data

Find Middle

**Read
Median**

This positional logic is exactly what makes the median robust. A value can move from 99 to 9,900 and, as long as it stays on the same side of the middle, the median does not move at all.

Median — Worked Calculation

Meridian Retail order values (₹ thousands), sorted: 12, 14, 15, 15, 18, 20, 22, 25, 99

With Bulk Order (n = 9, odd)

Position = $(9 + 1) / 2 = 5\text{th value}$

Median = 18

The 5th value in the sorted list is 18. Notice that 18 is below the mean of 26.67 — a gap that is already telling us about the right tail.

Without Bulk Order (n = 8, even)

Average of 4th and 5th values = $(15 + 18) / 2 = 16.5$

Median = 16.5

Statistic	With Outlier	Without Outlier	Change
Mean	26.67	17.63	+51%
Median	18.00	16.50	+9%

Robustness: The Breakdown Point

The breakdown point is the **proportion of data you can corrupt before a statistic becomes arbitrary**. It is the cleanest way to compare the robustness of two statistics.

Mean — Breakdown Point: 0%

One single value pushed to infinity takes the mean with it. The mean has no resistance whatsoever. This is not a flaw in the formula — it is the algebraic consequence of using every data point equally.

Replace 99 with 9,900 in Meridian: Mean becomes 1,117. The mean is now reporting a number larger than 8 out of 9 actual transactions by a factor of 60.

Median — Breakdown Point: 50%

You must corrupt **half the dataset** before the median moves arbitrarily. As long as fewer than half the observations are corrupted, the median holds its position.

Replace 99 with 9,900 in Meridian: Median stays at **18**. Unchanged. The median does not care how large the largest value becomes, only that it is still on the right side of the middle.

This is why regulators and statistical agencies report **median household income** rather than mean household income. The robustness is the point, not a technical preference.

Percentiles and Quartiles

The median generalises. Percentiles let you describe **any position in the distribution**, not just the midpoint. This is the foundation of all percentile-based reporting in analytics.

Definitions

P_k — the value below which k% of observations fall.

Q1 = P25 (lower quartile)

Q2 = P50 = Median

Q3 = P75 (upper quartile)

IQR = Q3 – Q1 — the range covering the middle 50% of the data.

Resistant for the same reason the median is: it ignores both tails entirely.

Meridian Five-Number Summary

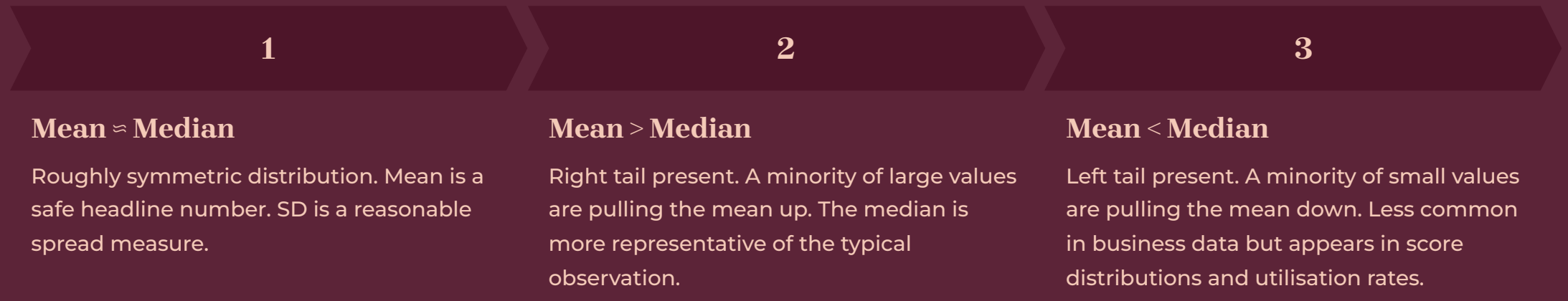
Min: 12 **Q1:** 15 **Q2:** 18 **Q3:** 22 **Max:** 99

IQR = 22 – 15 = 7

Notice the gap between Q3 (22) and Max (99). That gap — from the 75th percentile to the maximum — is more than 4 times the entire IQR. This is visible evidence of a tail before you compute a single additional statistic.

The Mean–Median Gap Is a Diagnostic

Compare the two before you compare anything else. The relationship between mean and median tells you the shape of the distribution at a glance — before computing skewness, before plotting a histogram.



- Meridian: mean 26.67 vs median 18.00. The mean sits **48% above the median**. That is a strong right tail, established in one subtraction. This is the first thing to check on any new numeric column — it costs two function calls.

Median in Production Reporting

Percentile reporting is standard practice wherever **tail behaviour is the business problem**. In each of these domains, the mean hides exactly what the audience needs to know.

Engineering SLAs

Teams report **p50, p95, p99 latency** — never mean latency. Mean latency hides the requests that lose customers. The p99 is the number that determines whether your SLA is achievable.

Compensation Benchmarking

Median salary by band and geography is the standard. Mean salary is distorted by a handful of very senior or very highly paid roles within each band.

Delivery Operations

The **p90 fulfilment time** is the promise to the customer. The mean is not — because the tail of slow deliveries is what customers actually experience and complain about.

Real Estate Indices

Median transaction price per sq ft by locality is the standard index measure. A handful of luxury transactions would distort a mean-based index beyond usefulness.

The Mode and Bimodality

The most frequently occurring value. The **only measure of centre valid for nominal data** — most common payment method, most common failure code, most common shipping city.

When to Use Mode

- Categorical and nominal data — mode is the only valid option
- When you want to identify the most common outcome, product, or category
- When a distribution is multimodal and you want to identify its peaks

A distribution can have one mode, two (**bimodal**), or many.

Bimodality is not a statistical nuisance — it is a signal.

Bimodality: A Critical Signal

Two peaks in a histogram mean you have two populations stacked in one column. Two customer segments, two pricing tiers, two shifts in a factory, two product categories incorrectly merged.

If a histogram shows two peaks, **stop computing summary statistics** and split the data first. Every statistic you compute on a mixed population describes a group that does not exist.

Choosing a Measure of Centre

The right choice depends on the data type, the shape of the distribution, and the audience. This reference table covers the most common situations you will encounter in practice.

Situation	Use	Reason
Symmetric, no extreme values	Mean	Uses all information, feeds further analysis cleanly
Long tail or outliers present	Median	Resistant to extremes; represents the typical observation
Categorical / nominal data	Mode	Only valid option for this data type
Two visible peaks in histogram	Neither — segment first	Summary statistics on mixed populations describe no one
Components of unequal size	Weighted mean	Equal weighting distorts the aggregate
Public audience or policy reporting	Median, or both	Mean invites misreading when the tail is visible

📌 When in doubt: report both and let the gap speak. A mean of 26.67 next to a median of 18 tells more of the story than either number alone.



PART 3 — STANDARD DEVIATION

Spread, in the Original Units

Same Centre, Different Business

Two delivery hubs. Both report a mean fulfilment time of 50 minutes. Are they equivalent operations? Not remotely.

Metric	Hub A	Hub B
Recorded times (min)	48, 49, 50, 51, 52	20, 35, 50, 65, 80
Mean	50.0	50.0
Standard Deviation	1.58	23.72

Hub A

An SD of 1.58 minutes means deliveries cluster tightly around 50 minutes. **Hub A can confidently promise 55 minutes** and will almost always deliver within that window.

Hub B

An SD of 23.72 minutes means deliveries range from 20 to 80 minutes. **Hub B cannot promise anything meaningful.** Identical centre, completely different operations — and the mean cannot tell them apart.

📌 **Centre without spread is not a description. It is a single point.** You need both to understand what is happening.

Range and IQR as Spread Measures

Range: The Crudest Measure

Range = Maximum – Minimum

Meridian: $99 - 12 = \mathbf{87}$

- Uses exactly **two observations** and discards the other $n - 2$
- Grows mechanically as sample size increases — useless for comparing datasets of different sizes
- Zero breakdown point — defined entirely by the two most extreme values
- Useful as a quick **sanity check on data entry**. Not useful as a reported statistic in any serious analysis

IQR: The Resistant Alternative

IQR = Q3 – Q1 — the width of the middle 50% of the data.

Meridian: $22 - 15 = \mathbf{7}$

- Resistant, for the same reason the median is: it ignores both tails entirely
- Pairs naturally with the median in all reporting
- **Standard outlier rule:** flag any value below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$
- Meridian upper fence: $22 + 1.5(7) = \mathbf{32.5}$ → the order of 99 is flagged automatically

This is the rule the box plot draws. The whiskers are the fences; the dots beyond them are flagged observations.

From Deviations to a Measure of Spread

We want to measure the average distance from the centre. The most direct route fails — and understanding why reveals the logic behind variance and standard deviation.

Meridian deviations from $\bar{x} = 26.67$:

-14.67, -12.67, -11.67, -11.67, -8.67, -6.67, -4.67, -1.67, +72.33

Their sum is exactly zero. Always. By the algebra of the mean. Positive and negative deviations cancel perfectly. We cannot average them to get a useful spread measure.

Option 1: Absolute Values

Take $|x_i - \bar{x}|$ for each observation, then average them. This gives the **Mean Absolute Deviation (MAD)**. Intuitive and easy to explain, but not differentiable — which makes it algebraically awkward for further analysis and optimisation.

Option 2: Square the Deviations

Square $(x_i - \bar{x})^2$ for each observation, then average. This gives **variance**. Harder to interpret directly (units are squared), but differentiable, additive across independent variables, and the foundation of essentially all inferential statistics.

Statistics chose squaring. That choice is why we then have to take a square root at the end to return to original units.

Variance

The mean of the squared deviations. A computational object more than a reporting one.

Sample Variance

$$s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$$

Used when you have a sample and want to estimate the population variance. Dividing by $n - 1$ is Bessel's correction — more on this shortly.

Population Variance

$$\sigma^2 = \Sigma(x_i - \mu)^2 / N$$

Used when you have the entire population. Divides by N , not $N - 1$, because no correction is needed.

Key Properties

Always ≥ 0 . Zero only when every value is identical. **Squaring makes large deviations dominate** — a value twice as far contributes four times as much variance. Units are squared (e.g. ₹² thousands²), which makes variance uninterpretable as a business number. It is a stepping stone, not a destination.

Standard Deviation — Definition and Meridian Result

$s = \sqrt{s^2}$ — the square root pulls the measure back into the original units of the data.

What It Tells You

Standard deviation is the **typical distance of an observation from the mean**, expressed in the same units as the data.

Meridian order values are in ₹ thousands, so s is in ₹ thousands. Directly comparable to the mean of 26.67.

$s = 27.44$ for Meridian (all nine orders).

What This Result Means

An SD of 27.44 against a mean of 26.67 means the typical deviation is nearly equal to the mean itself. In practical terms, an observation could reasonably be anywhere from near zero to over 50 — the SD is essentially telling you the data is **badly behaved**.

Same weakness as the mean, amplified — it is built on **squared deviations**, so a single extreme value inflates it heavily. The outlier of 99 dominates the SD calculation.

Standard Deviation — Worked Calculation

Clean subset for clarity, n = 5: **12, 14, 15, 18, 21**

01

Compute the Mean

$$(12 + 14 + 15 + 18 + 21) / 5 = 80 / 5 = \mathbf{16}$$

02

Compute the Deviations

$$(12-16), (14-16), (15-16), (18-16), (21-16) = \mathbf{-4, -2, -1, +2, +5}$$

Sum = 0. Confirmed: deviations always cancel.

03

Square the Deviations

$$16, 4, 1, 4, 25 \rightarrow \Sigma = \mathbf{50}$$

04

Divide

$$\text{Sample: } 50 / (5 - 1) = \mathbf{12.5} \mid \text{Population: } 50 / 5 = \mathbf{10.0}$$

05

Take the Square Root

$$s = \sqrt{12.5} = \mathbf{3.54} \text{ (sample)} \mid \sigma = \sqrt{10.0} = \mathbf{3.16} \text{ (population)}$$

Why $n - 1$, Not n ? (Bessel's Correction)

This question comes up in every statistics course and most job interviews. The answer is conceptually clean once you see it.

The Root Cause

Deviations are measured from $\bar{x}$, and $\bar{x}$ is computed from those same data points. The sample mean sits at the exact centre of the sample — it is literally the point that minimises the sum of squared deviations. So squared deviations from $\bar{x}$ are **systematically smaller** than squared deviations from μ would be.

The Consequence

Dividing by n **underestimates** the population variance. Dividing by $n - 1$ corrects for this bias. The $n - 1$ is called the **degrees of freedom**: once $\bar{x}$ is fixed, only $n - 1$ deviations are free to vary. The last one is entirely determined because they must sum to zero.

Practical Magnitude

At $n = 5$, the correction changes s by 12% — meaningful. At $n = 5,000$, the difference between n and $n - 1$ is 0.02% — irrelevant. Bessel's correction matters **for small samples**, which is precisely when you are most likely to be working from one.

Coefficient of Variation

Standard deviation cannot be compared across variables with different scales or units. A fund with an SD of ₹4% and a fund with an SD of ₹7% cannot be compared on absolute volatility alone — if they also have different expected returns, the comparison is meaningless.

CV = $s / \bar{x}$ (often reported as a percentage). CV normalises the spread by the level, making it a dimensionless ratio — comparable across variables, units, and scales.

Fund	Mean Annual Return	SD	CV (Risk per Unit of Return)
Fund A	8.0%	4.0%	0.50
Fund B	18.0%	7.2%	0.40

Fund B has nearly double the absolute volatility — but **less risk per unit of return**. Absolute SD would have picked Fund A as the safer choice. CV picks Fund B as the more efficient one. The difference matters for portfolio decisions.

 **Caveat:** CV is meaningless when the mean is near zero or the variable can take negative values. Do not apply it to temperatures, P&L, or net position data.

The Empirical Rule and Z-Scores

For an **approximately normal distribution**, standard deviation has a precise probabilistic interpretation via the empirical rule.

± 1 SD

68% of observations fall within $\bar{x} \pm 1s$.
About two thirds of your data.

± 2 SD

95% of observations fall within $\bar{x} \pm 2s$.
Covers virtually all typical variation.

± 3 SD

99.7% within $\bar{x} \pm 3s$. Extremely rare to see beyond this in a normal distribution.

Z-Score Formula

$$z = (x_i - \bar{x}) / s$$

How many standard deviations an observation sits from the mean.
Standardises any variable onto a common scale — useful for comparing across different variables.

The Meridian Warning

The outlier of 99 gives $z = (99 - 26.67) / 27.44 = \mathbf{2.64}$. It does not even clear the 3-sigma bar — because that outlier **inflated the very SD used to judge it**. On skewed data, z-scores are a weak outlier detector. Use the IQR fence ($Q3 + 1.5 \times IQR$) instead.



PART 4 — SKEWNESS

Is the Distribution Symmetric?

Shape Is the Third Question

Two datasets can share a mean and a standard deviation and still behave very differently, because the **asymmetry** differs. Centre tells you where values sit. Spread tells you how wide the distribution is. Neither tells you which side is longer — and that asymmetry determines where your risk actually lives.



Symmetric

Left and right tails mirror each other. The mean is a safe headline number. Most textbook examples assume this shape. Most real business data does not have it.



Right-Skewed

The long tail extends to the right. A minority of large values pull the mean above the median.

This is the default shape of commercial data — income, order value, latency, claim size.



Left-Skewed

The long tail extends to the left. A minority of small values pull the mean below the median. Appears in score distributions, utilisation rates, age at retirement.

Skewness measures the direction and degree of asymmetry in a single number. It also determines whether the mean is a safe number to report at all.

Skewness: The Formula and Its Logic

Skewness is built on the **third power** of the standardised deviations — and each of those design choices is intentional.

$$g_1 = [n / ((n-1)(n-2))] \cdot \sum [(x_i - \bar{x}) / s]^3$$

1

Why Cube?

Cubing **preserves sign** — negative deviations stay negative, positive stay positive. Squaring would make everything positive and destroy the directional information we want. The sign of the sum is decided by whichever tail is longer.

2

Why Does Distance Matter?

Cubing **amplifies distance** — a value 3s away from the mean contributes 27 units; a value 1s away contributes just 1 unit. Skewness is dominated by the far observations, not the typical ones.

3

Why Divide by s^3 ?

Dividing by s^3 makes the measure **dimensionless**, so it is comparable across variables measured in different units. A skewness of +2.87 means the same thing for order values in ₹ as for latency in milliseconds.

Right Skew (Positive): The Default of Commercial Data

The long tail extends to the right. Most values are small; a minority are very large. **Ordering: Mode < Median < Mean.**

Meridian: mode 15 < median 18 < mean 26.67. **Skewness = +2.87** — severely right-skewed.

Household Income

The median income in India's top decile is multiples of the national median. A small number of very high earners define the right tail.

Insurance Claims

Most claims are routine and affordable. A handful of critical illness or catastrophic damage claims hold the actuarial risk — and the average claim blends them unhelpfully.

API Latency

99% of requests resolve in under 200ms. A small fraction timeout at 10 seconds. Mean latency is around 230ms — meaningless for SLA design.

Order Value (Meridian)

Eight orders cluster between ₹12,000 and ₹25,000. One corporate order at ₹99,000 defines the tail. This is a structure, not an anomaly — it appears in every B2B retail dataset.

- ❏ Anything bounded below at zero and unbounded above will tend to skew right. This includes nearly all commercial and financial quantities.

Left Skew, Symmetry Thresholds, and Business Impact

Left Skew (Negative)

Ordering: Mean < Median < Mode

The long tail extends to the left. Most values are high; a minority are very low. Less common in business data because most quantities are bounded at zero.

Where it appears:

- Scores on an easy examination — most cluster near the ceiling
- Age at death in a developed economy — most cluster near 70–80
- Equipment utilisation in a well-run plant, capped near 100%
- Warranty replacement rates under a strict replacement policy

Reading the g_1 Value

Value of g_1	Interpretation
0	Perfectly symmetric
−0.5 to +0.5	Approximately symmetric — mean is safe
±0.5 to ±1.0	Moderately skewed — report median alongside mean
Beyond ±1.0	Highly skewed — mean is misleading on its own

Meridian, all nine orders: **+2.87**

Meridian, bulk order removed: **+0.52**

One observation moved the dataset from "roughly symmetric" to "severely skewed."

What Skewness Changes About Your Decision

Skewness is not a descriptive footnote. It changes the analysis workflow, the model choices, and how you handle outliers.

Reporting

Beyond $|g_1| > 1$, lead with the median. Report the mean only with the median beside it, and flag the direction of the tail explicitly. A mean of ₹26,670 next to a median of ₹18,000 is a complete picture; a mean of ₹26,670 alone is a misleading one.

Outlier Handling

Under strong right skew, high values may be **genuine population members**, not errors. A ₹99,000 corporate bulk order is a real transaction. Do not delete it by reflex. Decide whether it belongs to the same analysis question before removing it.

Modelling

Many regression and ANOVA methods assume roughly symmetric errors. Heavy right skew usually calls for a **log transform** of the target variable before modelling. Do not apply linear regression to raw order values with skewness above +1 and expect valid residuals.

Forecasting and Planning

A right-skewed cost distribution means the downside is bounded but the upside risk is not. **Plan the tail, not the average**. A revenue forecast built on the mean when skewness is +3 will be wrong in the worst direction during the worst quarters.



PART 5 — KURTOSIS

How Extreme Are the Extremes?

Tails Are the Fourth Question

Two distributions can be symmetric, share a mean, and share a standard deviation — and still differ in how often they produce extreme values. Standard deviation summarises typical distance. It says **nothing** about how often you get a value five standard deviations out.

Why Kurtosis Exists

In risk work, the difference between "a 5-sigma event happens once a decade" and "once a year" is the entire question — and standard deviation cannot answer it. Kurtosis measures the **weight of the tails relative to a normal distribution**.

A high-kurtosis distribution has more of its variance concentrated in the extremes. Most of the time, things are near the centre — but when they deviate, they deviate far.

The Formula

Raw kurtosis: $\beta_2 = \Sigma[(x_i - \bar{x})/s]^4 / n$

Excess kurtosis: $\beta_2 - 3$

The fourth power removes sign entirely — kurtosis reads both tails together. It amplifies distance harder than skewness: a value 4s away contributes 256 units.

The normal distribution has raw kurtosis of exactly **3**. Subtracting 3 sets the normal as the zero point. **Excess kurtosis is what almost every software reports** — pandas .kurt() and scipy.stats.kurtosis() both default to excess.

Three Tail Regimes

Type	Excess Kurtosis	Tail Behaviour
Leptokurtic	> 0	Fat tails — extreme values more frequent than in a normal distribution. Most of the time near the centre, but the rare events are really rare and really extreme.
Mesokurtic	≈ 0	Normal-like tails. The normal distribution is the reference. Financial models that assume this are accepting a simplification.
Platykurtic	< 0	Thin tails — extremes rarer than normal. Values are more uniformly spread. The uniform distribution is the extreme case.

Meridian — All 9 Orders

Excess kurtosis = **+8.42** — strongly leptokurtic. Fat tails driven entirely by the single observation of 99.

Meridian — 8 Orders (Bulk Removed)

Excess kurtosis = **-0.81** — mildly platykurtic. One observation removed swung the tail metric by more than 9 units.

The Lesson

Kurtosis is the most sensitive of the four statistics. It is driven by the most extreme values and reacts dramatically to a single tail observation. This sensitivity is the point — it is designed to catch tail risk.

Why Kurtosis Is a Risk Metric

Fat tails are where losses live. Kurtosis is the statistic that tells you whether your "rare event" assumptions are realistic.

Financial Returns

Equity returns are consistently leptokurtic. Models assuming normal tails — like classic Black-Scholes at the tails — systematically **underprice the probability of a large drawdown**. This mismatch was a significant contributor to the 2008 financial crisis.

Fraud and Anomaly Detection

High kurtosis in a transaction amount column indicates a heavy tail of unusual values. These are the observations worth investigating first. Kurtosis flags them as a class before you examine individual records.

Operational Risk

An outage duration distribution with fat tails means your average recovery time is not the number to staff against. The tail event — the 8-hour outage — determines your staffing and disaster recovery budget.

Data Quality Signal

Extreme kurtosis in a field that should be well-behaved (e.g. a product weight column) often signals **unit errors, duplicate entries, or a sentinel value** like 999 or -1 used as a missing data flag. Investigate before modelling.

The Peakedness Misreading — A Classic Error

The most common error in interpreting kurtosis appears in older textbooks and in interviews. It is worth addressing directly.

✗ The Wrong Reading

"High kurtosis means a tall, sharp peak in the middle of the distribution."

This interpretation was common in early 20th century statistics texts and has persisted through repetition. It is wrong, and it leads to incorrect analytical decisions.

✓ The Correct Reading

"High kurtosis means **heavy tails** — extreme observations occur more often than in a normal distribution."

The fourth power gives values near the mean almost no influence: a deviation of 0.2s contributes only 0.0016. The statistic is driven almost entirely by the far observations. Read kurtosis as: *how much of the variance is generated by extreme observations?*

A distribution can be completely flat in the middle and still have very high kurtosis, as long as its tails are heavy. The centre is almost irrelevant to the kurtosis calculation. This distinction matters when you are assessing tail risk, pricing derivatives, or investigating anomalies in operational data.

The Diagnostic Workflow

A repeatable order of operations for any new numeric column. Follow this sequence every time before drawing any conclusion.



Common Pitfalls to Avoid

- Averaging categorical codes (pin codes, product IDs)
- Reporting a mean on skewed data without the median beside it
- Using z-scores for outlier detection on skewed distributions
- Confusing raw kurtosis (normal = 3) with excess kurtosis (normal = 0)
- Summarising a bimodal column before segmenting the population
- Deleting flagged outliers before deciding whether they are errors or real

Quick Reference: All Formulas

Centre

Mean: $\bar{x} = \sum x_i / n$

Median: Middle value of sorted data

Weighted Mean: $\bar{x}_w = \sum (w_i \cdot x_i) / \sum w_i$

Spread

Range: max – min

IQR: Q3 – Q1

Sample Variance: $s^2 = \sum (x_i - \bar{x})^2 / (n - 1)$

Standard Deviation: $s = \sqrt{s^2}$

CV: $s / \bar{x}$

Z-score: $z = (x_i - \bar{x}) / s$

Shape

Skewness: g_1 based on $\sum [(x_i - \bar{x})/s]^3$

Excess Kurtosis: $\sum [(x_i - \bar{x})/s]^4 / n - 3$

Decision Thresholds

Skew $|g_1| > 1 \rightarrow$ report the median

Excess kurtosis $> 3 \rightarrow$ fat tails, treat as risk

Outlier lower fence: $Q1 - 1.5 \times IQR$

Outlier upper fence: $Q3 + 1.5 \times IQR$

Meridian Summary

n=9 values: 12, 14, 15, 15, 18, 20, 22, 25, 99

Mean = 26.67 | Median = 18 | SD = 27.44

IQR = 7 | Skewness = +2.87 | Excess Kurt = 8.42

Python Reference

All five statistics, in production-ready Python using pandas. One-liners for each.

```
import pandas as pd

df = pd.read_csv('orders.csv')

# Full descriptive summary: count, mean, sd, min, quartiles, max
df['order_value'].describe()

# Centre
df['order_value'].mean()
df['order_value'].median()
df['order_value'].mode()[0]

# Spread
df['order_value'].std()    # sample SD (divides by n-1)
df['order_value'].var()    # sample variance
df['order_value'].quantile(0.75) - df['order_value'].quantile(0.25) # IQR

# Shape
df['order_value'].skew()    # sample-adjusted skewness (g1)
df['order_value'].kurt()    # EXCESS kurtosis (normal = 0, not 3)

# Outlier fences (IQR method)
Q1 = df['order_value'].quantile(0.25)
Q3 = df['order_value'].quantile(0.75)
IQR = Q3 - Q1
lower_fence = Q1 - 1.5 * IQR
upper_fence = Q3 + 1.5 * IQR
outliers = df[(df['order_value'] < lower_fence) | (df['order_value'] > upper_fence)]
```

⚠ pandas .kurt() returns **excess kurtosis** (normal = 0). If you read a value and compare it to 3 as the baseline, you are reading raw kurtosis convention. These are different numbers. Always confirm which convention your tool uses before reporting.



The One Line to Take Away

**Never report a centre without a spread,
and never report either without
checking the shape.**

A mean of ₹26,670 is a data point. A mean of ₹26,670 with an SD of ₹27,440 is a warning. A mean of ₹26,670 with an SD of ₹27,440 and a skewness of +2.87 is a story — and that story tells you the mean itself is the wrong number to lead with.

The four questions — centre, spread, symmetry, tails — are not independent checks. They build on each other. Each one changes what the previous one means. Compute all four. Report the ones that are honest for the data in front of you.